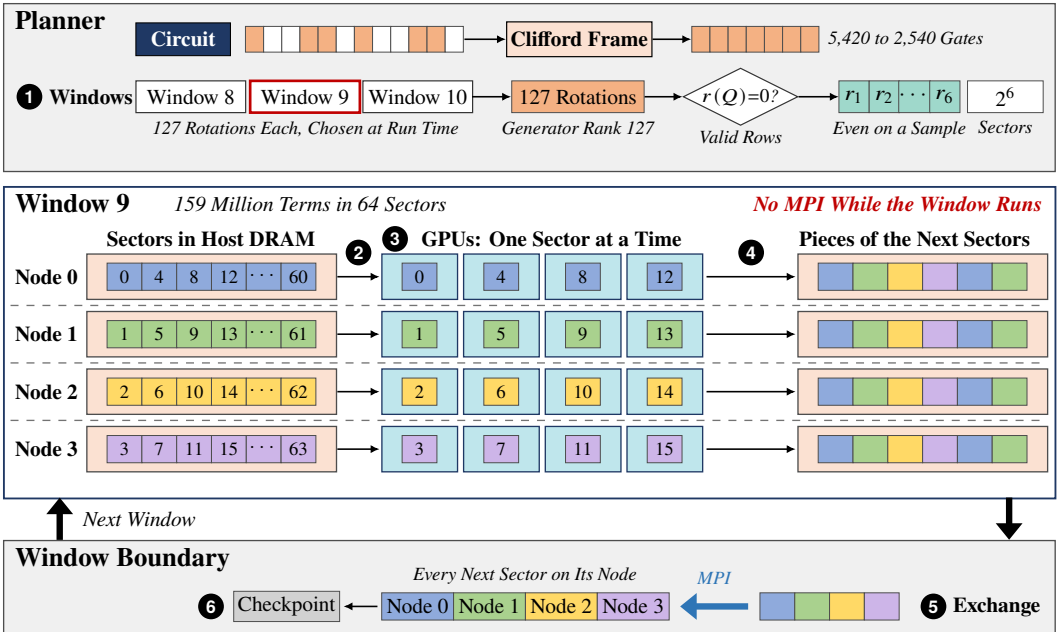

Planner
Circuit
Clifford Frame
5,420 to 2,540 Gates
1 Windows
Window 8
Window 9
Window 10
127 Rotations
r(Q)=0?
r1 r2 · · · r6
2^6
127 Rotations Each, Chosen at Run Time
Generator Rank 127
Valid Rows
Even on a Sample
Sectors
Window 9
159 Million Terms in 64 Sectors
No MPI While the Window Runs
Sectors in Host DRAM
2
3 GPUs: One Sector at a Time
4
Pieces of the Next Sectors
Node 0
0 4 8 12 · · · 60
Node 1
1 5 9 13 · · · 61
Node 2
2 6 10 14 · · · 62
Node 3
3 7 11 15 · · · 63
Next Window
Window Boundary
Every Next Sector on Its Node
MPI
6 Checkpoint
Node 0 Node 1 Node 2 Node 3
5 Exchange